Financial News Sentiment Prediction

Deep Learning Baseline Model Comparison Report

1. Project objective

The objective is to classify financial news text into three sentiment categories: **Bearish** (negative), **Bullish** (positive), and **Neutral** (neither clearly positive nor negative). Three recurrent neural network architectures were implemented using PyTorch: Simple RNN, LSTM, and GRU.

2. Dataset and preparation

Dataset detail	Observed value
Training CSV	9,543 rows before cleaning
Validation CSV	2,388 rows
Training rows removed	6 records with no meaningful text after cleaning
Final training rows	9,537
Text columns	text, label
Label mapping	0 = Bearish; 1 = Bullish; 2 = Neutral
Maximum sequence length	20 tokens
Vocabulary size	16,458 tokens (including PAD and UNK)

Preprocessing steps

URLs and user mentions were removed, extra whitespace was normalized, and text was lowercased. Text was tokenized with a regular expression that preserves common financial terms, numeric tokens, stock tickers, and simple smiley tokens. The vocabulary was built from training text only. Tokens were converted to integer IDs, then post-padded or truncated to 20 tokens. The supplied validation CSV was used; the accompanying XLSX was not used for model evaluation.

Class distribution in the final training data

Class	Count	Approx. share
Bearish	1,442	15.12%
Bullish	1,923	20.16%
Neutral	6,172	64.72%

The target distribution is imbalanced, with Neutral as the majority class. Macro F1-score is therefore reported alongside accuracy because it gives equal weight to the F1-score of each class.

3. Model configuration and results

All three models used an embedding dimension of 64, hidden dimension of 128, three output classes, batch size 64, and 10 training epochs. Adam was used with a learning rate of 0.001 and CrossEntropyLoss as the loss function.

Model	Accuracy	Macro F1-score
Simple RNN	72.91%	0.6008
LSTM	75.29%	0.6638
GRU	**76.68%**	**0.6743**

Result: GRU achieved the highest reported accuracy (76.68%) and macro F1-score (0.6743) among the three evaluation runs. LSTM ranked second, followed by Simple RNN.

4. GRU class-wise evaluation

Sentiment	Precision	Recall	F1-score	Support
Bearish	0.5507	0.5476	0.5491	347
Bullish	0.6395	0.5937	0.6157	475
Neutral	0.8483	0.8678	0.8580	1,566

GRU confusion matrix

Actual \ Predicted	Bearish	Bullish	Neutral
Bearish	190	49	108
Bullish	58	282	135
Neutral	97	110	1,359

5. Observations, limitations and conclusion

The GRU evaluation shows stronger performance on the Neutral class than on Bearish and Bullish. This may partly reflect the class imbalance. Training logs also showed training accuracy continuing to rise while validation loss increased in later epochs, indicating overfitting. The GRU's best validation accuracy during its recorded training run was 79.23% at epoch 6; its final-epoch accuracy was 76.68%. The LSTM and Simple RNN training runs also showed signs of overfitting.

Important evaluation note: The accuracy and macro F1 values in the model comparison table come from classification reports generated in separate evaluation runs. Each model was subsequently trained again when its weights were saved. Therefore, the table should be understood as a comparison of the reported evaluation runs, not as metrics measured from the exact saved checkpoints. For a strictly controlled final comparison, evaluate all saved checkpoints on the same validation set without retraining.

Dataset count note: The project brief reports record counts that do not match the supplied CSV files and are internally inconsistent. This report uses the actual supplied CSV counts and records the discrepancy rather than attempting to reconcile it without evidence.

Conclusion: Among the three baseline architectures, GRU achieved the strongest reported evaluation results. A Streamlit interface was implemented to accept financial news text, select one of the three models, and display a sentiment prediction with its model confidence. Three illustrative GRU tests returned the expected sentiment interpretations; these examples are functional checks, not a statistical performance measure.